Methodology

What each tool hunts for, why its filters are what they are, and how it is measured. One file, kept next to the code it describes.

Nine tools run the same software against different screeners. Each has its own database, its own model and its own training history — nothing is shared, so one tool's results cannot contaminate another's.

What applies to everything

The tradability floor

Three conditions are added to **every screener on every tool**, existing and future, at the moment a scan is sent. They are not in any screener's rule list because they are not in any screener — the scanner appends them on the way out.

Rule	Why
average volume ≥ 1M shares/day	Below this you cannot get out at a price you chose.
ATR ≥ $1	Below this there is not enough movement to pay for the risk.
ATR ≥ 3% of price	The same question relative to the stock. A $200 name with a $2 ATR clears the dollar test easily and still has only 1% of range to work with.

Editable in **Settings** — `minAvgVolume`, `minAtr`, `minAtrPct`. Setting one to 0 switches that leg off. Changing it changes every screener at once, which is the point of it living in one place.

How success is measured

There is **no stop, no target and no exit**. This is on purpose: the question being asked is "does this screener find stocks that move", which is the only thing a screener can be held responsible for. Whether a move is tradable is a question about setups, and comes after you know which screeners are worth keeping.

For every card, at the end of the day:

```
entry = the OPEN of the entry-minute bar
hh    = highest high from that bar to the close
ll    = lowest low  from that bar to the close

upR   = (hh − entry) / ATR14      how far it ran your way
downR = (entry − ll) / ATR14      how far it went against you first
```

Both are positive numbers. ATR14 is built from daily bars **strictly before today**, so nothing looks ahead. Bars are regular session only — pre-market and after-hours spikes cannot inflate a result.

The **opening range** (09:30–09:35 high and low) is recorded for every card on every tool, whether or not that tool trades it. The bars are already fetched, and those two levels are the trigger for the one day-trading setup with published evidence behind it — a month of data collected without them could not be asked the question afterwards.

Two more things are recorded on every card so the month can be asked about them later: **when it was found**, as minutes from the opening bell (negative before it), and whether **another tool had shortlisted it**. Both reach r1 and r4, so the scorecard and the model can use them.

> One caveat on the discovery time: cards captured before it existed have none, and the trainer fills a missing number with zero — which on this scale reads as “found exactly at the bell”. Only rows from before this change are affected, and they are a shrinking minority as the month fills in. Nothing else uses zero as a real value here.

A **good move** is `upR ≥ 1.3`. The same threshold the model calls a win, so the scorecard and the model are not grading on different curves.

Worked example — entry \$10, high \$12.50, low \$9.40, ATR \$1: **upR 2.5, downR 0.6.**

The learning moment

Each tool takes **one photo a day** of every card on screen, then measures how far each ran from two entry times shortly after. Photo plus result is one training row — the only thing the score model ever learns from.

A stock found after the photo still shows on screen and can be traded, but it teaches the model nothing. The photo time therefore follows each tool’s own session, and always lands one minute after one of that tool’s discovery scans, so it freezes a fresh registry rather than the previous scan’s.

Discovery vs refresh

A screener’s run window gates **discovery only** — whether new candidates are added. Cards already on screen keep re-quoting every five minutes from 04:00 to 16:00 regardless, because a card found at 09:40 is still being watched at 15:00 and its price, VWAP, distances and tags have to keep up with the tape.

Mirrors

Most tools ship a screener and its **mirror** — the same structural setup facing the other way — so a month of data can answer “is this edge directional, or does this screener just find movers?”.

Mirroring is not blind operator inversion:

- **Directional rules flip sign**: `change > 5` becomes `change < −5`.
- **High/low pairs swap to their counterpart**: `close ≥ 1-month high` becomes `close ≤ 1-month LOW`. Inverting only the operator would give “below the monthly high”, true of nearly every stock, screening for nothing.
- **Bounded oscillators reflect**: `RSI > 70` becomes `RSI < 30`, not `RSI < 70`.
- **Quality guards survive untouched**: `price > $1`, `volume > 2M`. Inverting a guard selects illiquid junk, not the bearish counterpart of the setup.

T1 — Screener · port 3000

The control. The original tool, and the only one with history: a month of cards, the model trained on them, and the one edge measured out of them so far.

Screeners — all three run all day

Trend — intraday trend continuation.

```
close ≥ 20                            established price, not a penny stock
close ≥ SMA5   and  close ≥ VWAP      above the short average and the day's average price
close|1W > VWAP|1W                    above the weekly VWAP
close|1M > VWAP|1M                    above the monthly VWAP
EMA50|1 > EMA120|1                    daily trend is up
close|1 ≥ EMA50|1                     holding the daily trend line
VWAP ≥ SMA75|5                        intraday VWAP above the 5-minute 75 average
relative_volume_intraday|5 > 3        volume arriving right now
relative_volume_10d_calc > 1.5        and elevated for the day
average_volume_90d_calc > 1M          liquid
```

Alignment across every timeframe at once — day, week, month, and the last few minutes.

Pre-Mkt — pre-market participation.

```
premarket_volume > 1.5M      real pre-market trade, not an indication
relative_volume_10d_calc > 3
average_volume_10d_calc > 2M
close ≥ 1
```

Big Move — the crude one, and it works.

```
relative_volume_10d_calc > 10     ten times normal volume
close ≥ 2
average_volume_10d_calc > 2M
```

Why T1 has no run windows

Pre-Mkt is built on `premarket_volume`, which stops moving at the bell — its filters plainly say "pre-market only". It is still left running all day, on purpose.

The one edge measured so far is the **overlap between Big Move and Pre-Mkt**: 54% of those cards made a good move against 13% for everything else (Fisher p = 0.0015, spread over 11 days, survives dropping the two biggest winners). That overlap is read off the cards frozen at 09:36. Giving Pre-Mkt the window its filters suggest would stop it running in the 09:30–09:36 scans and strip the Pre-Mkt tag off every frozen row — deleting the measurement rather than improving it.

T1 stays exactly as it was, so the other six have something to be compared against.

Capture: photo 09:36, entries 09:37 / 09:40.

T2 — Momentum · port 3010

Looking for: a stock in a clean daily uptrend breaking to a new monthly high.

```
SMA5|1 > EMA9|1 > EMA13|1 > EMA20|1     daily averages stacked fast-over-slow
close ≥ High.1M                          breaking the 1-month high
close ≥ 1
average_volume_10d_calc > 500K
```

Why these filters. The stack is a statement about the daily trend: each faster average above the slower one means every recent period has been stronger than the one before it. On its own that only says the stock has been rising — the monthly-high break is what makes it an event rather than a description.

The stack barely moves intraday, so what the screener is really timing is the break.

Mirror: stack inverted, `close ≤ Low.1M`. Breakdown from a downtrend.

Window 09:30–15:00. A break on pre-market liquidity is not a break, and one in the last hour leaves no session to work with.

Capture: photo 09:46, entries 09:47 / 09:51 — fifteen minutes past the open, so a break that was one opening spike has already failed and is not in the photo.

T3 — Gappers · port 3020

Looking for: a small-float stock gapping up on real pre-market volume.

```
premarket_change > 5                    gapping up at least 5%
premarket_volume > 500K                 people actually traded it
relative_volume_10d_calc > 2
close ≥ 1
float_shares_outstanding < 50M          small float
```

Why these filters. A gap with no pre-market volume is a quote, not a move — it can vanish at the bell. The float cap is the deliberate part: a small float is what lets a gap keep extending after the open, because there is not much stock available to sell into the buying.

Mirror: `premarket_change < −5`, everything else identical. Gap down.

Window 04:00–10:30. `premarket_change` and `premarket_volume` freeze at the bell — after that they are yesterday's numbers. Left running all day this screener would re-report the same names until the close and none of it would be new. It gets the pre-market plus the first hour, where a gap either continues or fails.

Capture: photo 09:36, entries 09:37 / 09:40 — a gap is an opening trade.

T4 — VWAP Reclaim · port 3030

Looking for: a stock that sold off, then pushed back up through VWAP with volume behind it. **The reversal tool.**

```
close crosses_above VWAP                the reclaim itself
relative_volume_10d_calc > 2            volume behind it, not drift
close ≥ 1
average_volume_10d_calc > 1M
```

Why these filters. VWAP is where the average buyer of the day is even. Crossing back above it flips that group from losing to winning, which is why reclaims tend to run. Without the volume condition the cross is drift, and drift across VWAP reverses as easily as it holds.

Mirror: `crosses_below`. The loss of VWAP.

Window 09:45–15:30. This is the tool that deliberately stays awake into the afternoon, because reversals come late — the "move first, then reverse" case. It starts at 09:45 rather than the bell because VWAP computed off the first few prints is not yet a level anything is reclaiming, and stops at 15:30 because a reclaim in the last half hour has no session left to resolve in.

Capture: photo 10:01, entries 10:02 / 10:06.

Known limitation: a cross is a moment, and discovery runs every 15 minutes after 10:00. Some crosses will be missed between scans.

T5 — 52-Week Break · port 3040

Looking for: an established company clearing its 52-week high.

```
close ≥ price_52_week_high        at or through the yearly high
market_cap_basic > $1B            established, not a small cap
relative_volume_10d_calc > 1.5
average_volume_10d_calc > 1M
```

Why these filters. The billion-dollar floor is the deliberate part, and it is there for a reason beyond the strategy: every other tool samples cheap small caps, which is what let share price dominate the factor model. This tool gives that finding something to be tested against.

At a 52-week high there is nobody holding the stock at a loss waiting to sell into strength — no overhead supply.

Mirror: `close ≤ price_52_week_low`. The yearly low.

Window 09:30–16:00. These break on institutional flow, which arrives at any hour and often late in the day. Pre-market is excluded: a 52-week high printed on a handful of thin shares is not a break.

Capture: photo 09:46, entries 09:47 / 09:51.

T6 — Overextended · port 3050

Looking for: a stock stretched far enough to be worth watching for a snap back. **The fade tool.**

```
RSI > 70                          overbought
close ≥ EMA20                     extended above its own recent average
relative_volume_10d_calc > 3      the extension is being driven, not drifting
close ≥ 1
average_volume_10d_calc > 1M
```

Why these filters. RSI alone finds stocks that have been strong for weeks without going anywhere today. Pairing it with the volume condition restricts it to extension being made right now.

This tool exists to answer a question rather than to state a view: **does an extreme continue, or does it revert?** The mirror is what makes that answerable.

Mirror: `RSI < 30`, `close ≤ EMA20`. Genuinely oversold.

Window 10:00–16:00. At the open RSI still describes yesterday’s close; extension is something the session builds. Then it runs to the close, because a stock can be stretched at any hour and the fade is the trade.

Capture: photo 10:16, entries 10:17 / 10:21.

T7 — Liquid Movers · port 3060

Looking for: heavy real volume and real range, split by session. The only tool that refuses thin stocks outright rather than as a side effect.

Its two screeners are **not** a mirror pair. Each is a complete setup for its own session, and the sessions do not overlap, so a stock lands in one or the other.

Pre-Market Gap — 04:00–09:30

```
gap not between −3% and +3%          moving either way, at least 3%
ATR ≥ $1
average_volume_90d_calc ≥ 2M
```

Direction-agnostic on purpose. TradingView has no absolute-value operator, so “gap up or down 3%” is written as “outside −3%…+3%”, which is the same set.

After Open Volume — 09:30–16:00

```
relative_volume_10d_calc > 3
volume > 10M shares                 actually traded today, not an average
close > 1
```

Why these filters. `volume` rather than average volume is the distinctive choice: it asks what has traded *today*, so the screener only fires once real participation has shown up.

Capture: photo 09:36, entries 09:37 / 09:40. The pre-market screener is what feeds training here — its candidates are all in by the bell. The after-open screener finds tradable stocks all day, but they arrive after the photo and do not train the model.

T8 — CANSLIM · port 3070

Looking for: O'Neil's growth criteria — a profitable, accelerating company at a new high while leading the market. Deliberately slow: a handful of names that stay interesting for months, not a daily hunt.

```
earnings_per_share_diluted_yoy_growth_fq ≥ 25     C — latest quarter, YoY
earnings_per_share_diluted_yoy_growth_fy ≥ 25     A — and not a one-quarter accident
total_revenue_yoy_growth_fq ≥ 15                  sales behind the earnings
close ≥ price_52_week_high                        N — at or through the 52-week high
Perf.6M > 30                                      L — leading
relative_volume_10d_calc > 1.5                    S — demand showing up today
total_shares_outstanding_fundamental < 1B         S — a cap on supply
market_cap_basic > $300M                          institutions cannot buy what does not
trade
```

Second screener — CANSLIM Pullback. Same company quality, but back at its 50-day rather than breaking out: `close ≥ SMA50` and `close ≤ EMA20`. O'Neil buys breakouts; the pullback is where the same name is usually cheaper. It is not a mirror — the mirror of a growth screener would be a collapsing company, which is not what this tool is for.

What CANSLIM cannot be, honestly

Letter	Status
I — institutional sponsorship	**Left out.** There is no dependable screener column for it. Faking it with something that merely sounds similar would be worse than its absence.
M — market in an uptrend	**Not a filter, on purpose.** It describes the whole market, not a stock. The regime engine already stamps it on every card; filtering on it here would hide candidates on weak days — exactly the days worth recording for the comparison later.
L — leader	**The weak link.** O'Neil's RS Rating is a percentile rank against every other stock, which a screener cannot compute. Six-month performance is a plain threshold, so a strong market lets more names through and a weak one fewer. It measures strength, not leadership.

The cross-tag

CANSLIM matches are written to a shared list every other tool reads. When one of those names turns up in an unrelated screener — a VWAP reclaim, a gap, a big volume day — the card is tagged ★ **CANSLIM** with how many days it has held its place on the list.

Membership lasts **90 days** from the last time a name qualified, so it does not drop off the day it stops printing a new high. Re-qualifying extends it without resetting its age.

This is the only place tools see each other, and the shape of it matters: the list is a **label, never a filter**. Reading it cannot add or remove a single candidate from any screener, and a tool that cannot find or parse the file scans normally with no tags. The isolation that matters — one tool's data never deciding another tool's candidates — is intact.

`canslim` also travels into the registers as a yes/no field, so the model can learn from it and the scorecard can be sliced by it.

Capture: photo 09:46, entries 09:47 / 09:51 — the same timing as T2, so the two breakout tools stay directly comparable.

T9 — Stocks in Play · port 3080

The benchmark, and deliberately the dumbest screener here.

```
close > 5
```

That is its only rule. The tradability floor supplies the rest — average volume ≥ 1M, ATR ≥ $1, ATR ≥ 3% of price. Sort by relative volume, take the top 20.

No pattern. No direction. No structure. Just *what is unusually active today*.

Why it exists. Every other tool makes the same implicit claim: that some structure — a stacked moving average, a VWAP reclaim, a 52-week break — finds better movers than simply taking today's most active liquid names. Until this tool there was nothing plain to test that claim against, because every tool was clever.

At the end of the month the comparison reads directly:

```
T9  Stocks in Play      35% good moves     ← the number to beat
T2  MA Stack Breakout   52%                the structure is earning its keep
T4  VWAP Reclaim        34%                no better than doing nothing clever
```

It is not a straw man. In the one strong published day-trading result, the edge came almost entirely from the relative-volume screen: dropping it took the same strategy from a Sharpe of 2.81 to 0.48, below buy-and-hold. Ranking by unusual volume is a hard benchmark.

No mirror. There is no direction to flip. The opposite of "unusually active" is "quiet", which is not the other side of this setup — it is the absence of it.

Nothing structural may ever be added to it. The moment this screener says something about trend or pattern it becomes another clever tool, and there is nothing plain left to compare the clever ones against. A test enforces this.

Capture: photo 09:36, entries 09:37 / 09:40 — identical to T1 on purpose. A different capture time would make every comparison partly a comparison of clocks rather than of screeners.

The unified shortlist

Every tool keeps its own shortlist — each is a separate experiment. But the trader is one person with one account, so **anything shortlisted on any tool appears on the unified list, by design**. There is nothing to opt into.

It is on the landing page, sorted by how many tools agreed. A name three tools picked independently is a different proposition from one that appeared on a single list, and that is the question the list exists to answer.

Each tool publishes its own shortlist whenever it changes — the auto-rule, a manual star, an export — so the union is never behind. On a card, a name another tool has starred shows ★ **also T1 T7**. That badge never touches this tool's own star: `inShortlist` stays this tool's decision, and it is the one the model reads.

Same rule as the CANSLIM share: a **view, never a filter**. Reading it cannot add or remove a candidate, change a score, or alter a register row anywhere.

Reading the scorecard

Analysis → Screener scorecard, per screener, over every captured day:

Column	Meaning
Cards / Days	sample size — under 20 cards or 5 days is left unjudged
Median R	half its cards did better than this
Avg R	average move in your favour
Avg −R	average move against you first (stored positive)
≥1.3R	share of cards that made a good move
≥2R	share that made a big one
Days hit	share of days that produced at least one good move
Verdict	keep / watch / drop

Two comparisons, on the landing page

Compare every screener pulls all nine scorecards into one table and ranks every screener against **Stocks in Play** — the plain list. A screener has to beat it by 5 points to count, because a month gives a few dozen cards each and two or three points on that sample is noise. Entry A and entry B are both selectable.

Its limit, stated plainly: the plain list samples liquid stocks over $5, and some tools deliberately sample elsewhere — small floats, billion-dollar companies — so part of any gap is a difference of universe rather than of screener. It is still the right question, because the alternative is trading a screener with nothing to compare it to; just do not read a 3-point gap as proof.

Does direction matter? pairs each screener with its own mirror. This is the cleaner of the two, and the reason the mirrors exist: a screener and its opposite-facing twin run in the same window over the same universe by construction, so what separates them is direction and little else. Three outcomes:

Verdict	Meaning
directional	one side clearly beats the other — the direction is doing real work
both sides work	it finds movers; the direction is decoration. Worth knowing before trading it one way.
neither beats the plain list	the pair is noise whichever way it faces

Days hit is the column that decides. A screener whose good moves all landed on two wild days has a flattering average and is not something anyone can plan around, so a *keep* needs both a rate and a spread of days.

Each screener is compared against **its own tool's** pooled baseline. Comparing across tools compares universes — one samples two-dollar small caps, another samples billion-dollar companies — not screeners.

Order of work

1. **Collect.** Run the tools for about a month. Nothing to decide yet.
2. **Cut.** Read the scorecard. Delete the screeners that do not find movers.
3. **Then setups.** Only for the screeners that survived, work out which entry, stop and target suit what they find. That is the question the current measurement deliberately does not answer.